Clinton Enwerem

United States

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Summary

Robotics Research Engineer with 5+ years of research experience and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Develop demonstration data pipelines, evaluate imitation and reinforcement learning policies, and implement Python/C++ sensing and control software for dexterous manipulation in simulation and on physical, poly-articulated robots.

Skills

Robotics: ROS/ROS 2, ros2_control, MoveIt 2, MoveIt Servo, SocketCAN, VR teleoperation; **Planning & Control:** RRT*, FCL, state estimation, CBF-CLF QPs, MPC; **Learning:** PyTorch, OpenVLA-7B, QLoRA, ACT, DAgger, Diffusion Policy, PPO, flow matching; **Simulation:** Isaac Lab/Sim, Gazebo, MuJoCo, Drake; **Software:** Python, C++, Linux, Docker, git, CI; **GPU Computing:** MuJoCo Warp, JAX; **Perception:** FoundationPose, BundleSDF, SAM 2, SAM 3D, PCL, OpenCV; **Robots:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, RealHand A7 (Bimanual Arm-Hand System), RealHand L6, LEAP Hand v1, Unitree G1 (simulation).

Work Experience

Institute for Systems Research (ISR), University of Maryland Aug. 2021 – Present
Graduate Research Assistant *College Park, MD*

• Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under contact uncertainty; implement Python/C++ and ROS 2 software for physical arm–hand systems.
• Trained PPO with GDF rewards in **1,024 Isaac Lab environments**: **64/64** pregrasp arrivals versus **0/64** with palm-distance rewards. Published distributional RL (**CDC 2025**): **448 goals** versus **329/239** across **100 MuJoCo episodes**, with similar constraint-violation costs.
• Developed VR teleoperation software for an xArm7-RealHand-L6 robotic manipulator with wrist-to-Cartesian-velocity mapping, relative-joint-position hand control, grasp primitives, thumb-abduction control, and hand-state reset; integrated MoveIt Servo with Dimensional OS (DimOS) for real-time collision detection, arm speed modulation, and velocity damping near kinematic boundaries.
• Fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing; each of three training seeds achieved 136/200 task completions versus 129/200 without supervision, recovering seven failures without disrupting nominal successes.
• Built depth-to-trajectory planning with RANSAC/DBSCAN segmentation, point-to-plane ICP registration, and constrained CasADi/IPOPT optimization; verified kinematics, joint limits, and path constraints in MuJoCo.

MATRIX Lab, USM at Southern Maryland Jun. – Aug. 2022
Research Intern *California, MD*

• Identified a discrete-time Crazyflie model from flight data and implemented a finite-horizon optimal formation controller in Python and Crazyswarm; executed leader–follower formation flights with 10 physical Crazyflie quadrotors under localization uncertainty.

Kognitive Robotics Mar. 2021 – Aug. 2021
Robotics Engineer *Lagos, Nigeria*

• Designed power and sensor–actuator interfaces in Altium; assembled a mobile robot and developed and tested its ROS software.

Robotics & Artificial Intelligence Nigeria (RAIN) Jan. 2020 – Feb. 2021
Robotics Trainee *Ibadan, Nigeria*

• Developed a ROS-based quadrotor autonomy stack on a Raspberry Pi 3, integrating ORB-SLAM3 visual SLAM, an EKF fusing SLAM poses with IMU data, and a PID flight controller. The onboard controller maintained stable hover during ground-station packet loss.

Projects

ParcelStow — Execution-Timing Robustness in Imitation Learning (Paper, project, Code, Data & Models) | *Isaac Lab, PyTorch* 2026
• Built ParcelStow, an Isaac Lab benchmark for assessing imitation-policy robustness to execution-timing variation; generated **970K** expert states (**297** demonstrations) and trained ACT, Diffusion Policy, and DAgger. Controlled timing stress tests exposed precision-insertion failures: at 2× execution speed, ACT achieved **53%** success versus **84%** for the expert.

Grasp Distance Fields — Feedback Grasp Execution (Project, Paper) | *C++, Python, QP* 2026
• Implemented a CBF-CLF quadratic-program controller for feedback grasp execution; simulated experiments lifted **46 of 50** objects, with a median QP solve time of **0.09 ms**.

realhand_l6_ros2 — Dexterous-Hand Control Software (Code) | *C++, SocketCAN* 2026
• Implemented joint/tactile-state interfaces, contact-gated control, and MoveIt 2 configuration with mock and virtual-CAN CI tests. Verified communication, feedback, tactile acquisition, and trajectory execution on a physical RealHand L6.

Selected Publications

• **C. Enwerem et al.** "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds." *arXiv preprint*, 2026. arXiv:2609.01453, project page, code, data & models.
• **C. Enwerem et al.** "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
• **C. Enwerem et al.** "Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression." *CDC 2025*. arXiv:2506.06954.
• Additional first-author papers in control, safe learning, and robotic manipulation: IROS 2026; CDC 2026/2025/2024. Full list: Google Scholar.

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. Aug. 2018